

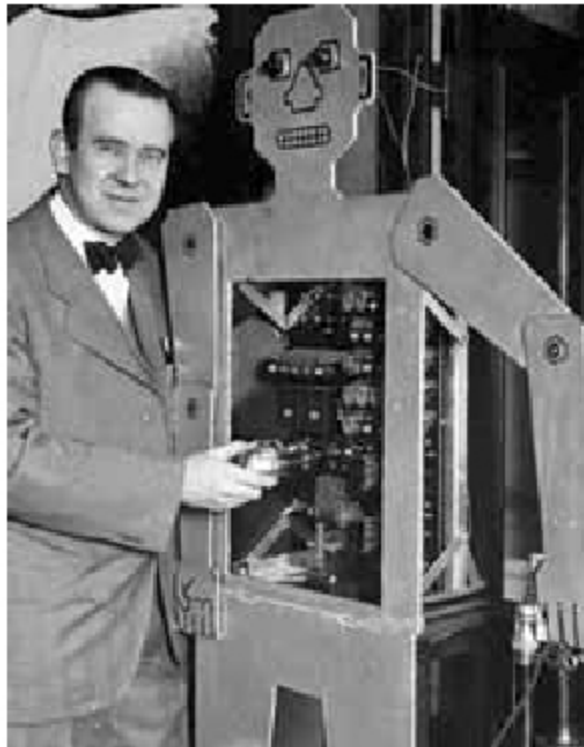


Fig. 2: Herbert Televox humanoid robot (Credit: www.lucarobotics.com)



Fig. 4: Sophia (Credit: en.wikipedia.org)

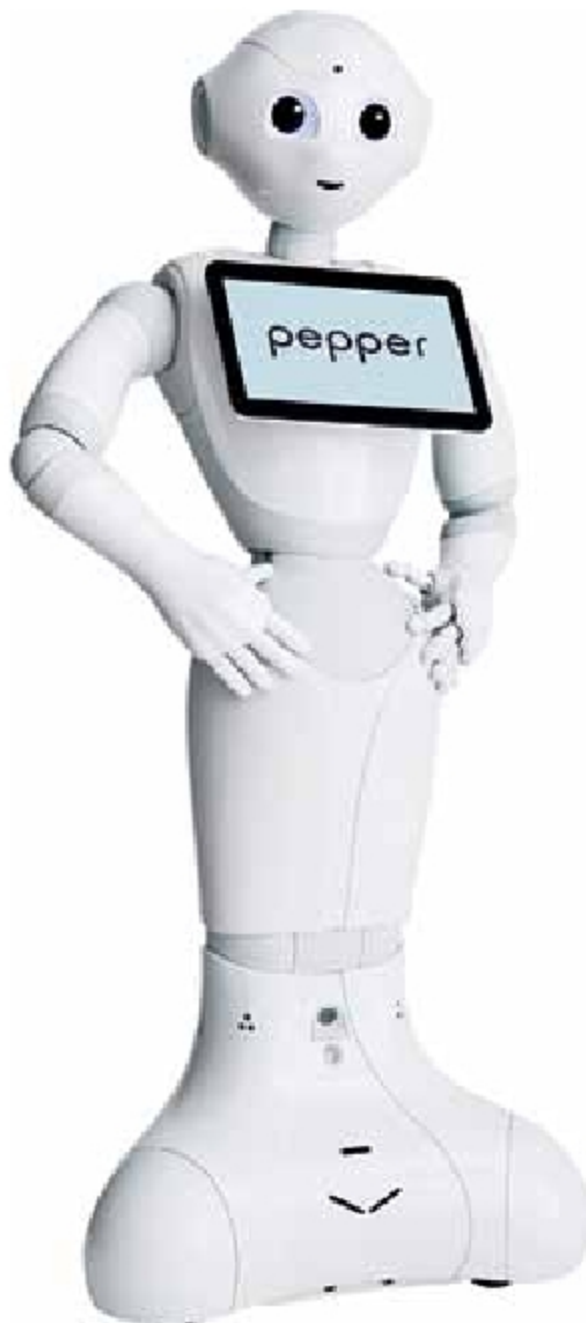


Fig. 3: Pepper robot (Credit: www.softbankrobotics.com)



Fig. 5: AI news anchor (Credit: <https://newsroompost.com>)

1. First humanoid robot

Herbert Televox (Fig. 2) was the first humanoid robot built by Ron Wensley in 1927. The robot could lift the handset to receive a telephone call and control simple processes by operating switches according to the signals it received. While the robot could not speak when it was first created, it later learnt to say two simple sentences. This was the beginning and starting point for the developmental process.

2. First social humanoid robot

Pepper (Fig. 3) is the world's first social humanoid robot that can recognise faces and understand basic human emotions. It was manufactured by SoftBank Robotics and introduced in 2014. Pepper can engage with people through conversations and touchscreen on the chest. The touchscreen on its chest displays messages. It also has a speech recognition feature and can speak in fifteen languages, including English, French, Spanish, German, Italian, Arabic, and Dutch.

3. First humanoid robot citizen

Sophia (Fig. 4) was the first robot to receive citizenship in a country (Saudi Arabia) in 2017. It was developed by Hong Kong-based company Hanson Robotics in 2016. It imitates human gestures and facial expressions, and is able to answer certain questions and make simple conversations on predefined topics. In 2020, Sophia visited the City of Joy, Kolkata, after touring 65 countries. This was Sophia's second visit to India. She first visited India in 2017.

4. First news anchor AI-robot

Qiu Hao, a lifelike image male news anchor (Fig. 5), was developed in 2018 by Xinhua and the Chinese search engine Sogou. It uses machine learning (ML) to simulate the voice, facial movements, and gestures of real-life broadcasters. It can work continuously and tirelessly. This AI technology is not only limited to news presenting but can also be customised to different applications in other industries.

5. First robotic avatar

FEDOR and T-HR3 (Fig. 6) were among the first avatars introduced in

lion dollars by 2026 at a CAGR of 36.4 per cent, increasing from 803.7 million dollars in 2018. The global market has been dominated by the North America region and is projected to account for 3,052.7 million dollars at the end of 2026.

Some of the key players in the humanoid robot market include SoftBank (Japan), ROBOTIS (South Korea), Honda Motor (Japan), UBTECH Robotics (China), Hanson Robotics (Hong Kong), PAL Robotics (Spain), Toyota Motor (Japan), Istituto Italiano di Tecnologia (Italy), Engineered Arts (UK), Robotics Lab (Spain), and National Aeronautics and Space Administration (NASA, US).

This article covers ten humanoid robots that would potentially revolutionise the robotics industry.

shipments are expected to reach one million units by 2024.

In another report by Research Dive, the global humanoid robot market is expected to reach 9,630 mil-